A Global Hamiltonian Period Atlas for Positive Lotka–Volterra Flow

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Abstract

We give a global, source-native description of the positive two-species Lotka–Volterra flow at its physical time clock. A logarithmic normalization turns the system into a strictly convex proper Hamiltonian flow. Every positive energy is therefore one compact periodic oval. The two real Lambert-W branches of the exponential potential give exact one-dimensional quadratures for the enclosed area and period, and coarea gives the action identity $J'(h) = T(h)/(2\pi)$. Linearization supplies the center-period limit, while integrating the log equations gives exact prey and predator cycle averages. An independent certificate checks 24 levels in six rational parameter sets and re-integrates the period with a SciPy event ODE. We do not assert period monotonicity or a high-energy asymptotic, and we make no target arithmetic or operator claim.

1 Model and global claim

Consider

$$\dot{x} = x(a - by), \qquad \dot{y} = y(-c + dx), \qquad a, b, c, d > 0,$$

on $x, y > 0$, with physical time t. Put $x_* = c/d$, $y_* = a/b$ and $u = \log(x/x_*), v = \log(y/y_*)$. Then

$$\dot{u} = a(1 - e^v), \qquad \dot{v} = c(e^u - 1), \qquad H = c(e^u - u - 1) + a(e^v - v - 1).$$

Theorem 1 (period annulus and exact atlas)**.** *H is a strict convex proper first integral with unique critical point $(0, 0)$. Every level $H = h > 0$ is a smooth compact periodic oval. If $F(s) = e^s - s - 1$, let*

$$\ell(r) = -W_0(-e^{-1-r}) - 1 - r, \qquad q(r) = -W_{-1}(-e^{-1-r}) - 1 - r.$$

With $u_- = \ell(h/c)$, $u_+ = q(h/c)$, and $r(u) = (h - cF(u))/a$, its area and period are

$$A(h) = \int_{u_-}^{u_+} \left[q(r(u)) - \ell(r(u))\right] du,$$

$$T(h) = \frac{1}{a} \int_{u_-}^{u_+} \left[\frac{1}{1 - e^{\ell(r(u))}} + \frac{1}{e^{q(r(u))} - 1}\right] du.$$

The endpoint singularities are integrable. For $J(h) = A(h)/(2\pi)$, $J'(h) = T(h)/(2\pi)$, and $\lim_{h \downarrow 0} T(h) = 2\pi/\sqrt{ac}$. Every periodic orbit obeys $\langle x \rangle = c/d$ and $\langle y \rangle = a/b$.

Proof. The Hessian is $\operatorname{diag}(ce^u, ae^v)$ and F tends to infinity at both ends, giving strict convexity and properness. The sublevel $K_h = \{H \leq h\}$ is a compact convex body with nonempty interior, so its boundary $H = h$ is connected; regularity follows because the only critical point is the origin. The canonical equations are $\dot{u} = -H_v, \dot{v} = H_u$, and the vector field is nonzero on this boundary. Thus the connected compact one-manifold is one periodic oval. Solving $F(s) = r$ gives the two displayed Lambert branches. Splitting the oval into lower and upper graphs gives the quadratures. Along the level, $|\dot{(u, v)}| = |\nabla H|$, hence $dt = ds/|\nabla H|$; the coarea formula then gives $dA/dh = \int_{H=h} ds/|\nabla H| = T$. The center matrix is $\left(\begin{smallmatrix} 0 & -a \\ c & 0 \end{smallmatrix}\right)$. Finally, integrating $\dot{u}$ and $\dot{v}$ over a period gives $\langle e^u \rangle = \langle e^v \rangle = 1$. $\square$

2 Branches, action, and boundaries

The branch labels matter: W_0 gives the negative graph and W_{-1} the positive graph. A trigonometric substitution in the certificate cancels the square-root endpoint singularities without changing the physical clock. The geometric action is positive by definition; no orientation convention is needed. The axes are invariant one-dimensional flows, not members of the positive period annulus. If a rate is zero, coercivity or the interior Hamiltonian statement can fail and the resulting boundary equation is treated separately. We intentionally leave period monotonicity and large-energy asymptotics outside the theorem.

3 Independent audit and Route-A boundary

The released ledger contains six positive rational (a, b, c, d) cases and four energies per case. It serializes 24 areas, periods, actions, branch endpoints, and residuals. The producer uses high-precision Lambert-W quadrature; the checker independently repeats those identities and then integrates the log flow with a DOP853 event solver from the right turning point. The checker passes 732 assertions, including all 24 heterogeneous center-period checks; SymPy passes 12 identities, byte replay is exact, and twelve repaired/stale hash attacks are rejected.

This is a continuum of real-energy cycles, not a discrete arithmetic primitive ledger. Accordingly the Route-A tuple is

`(A0_FAIL, A1_WEAK, A2_FAIL, A3_FAIL, A4_FORMAL_HINT),`

with `overall=ROUTE_A_REJECTED` and `route_b_invocation_allowed=false`. No target prime, zero, local factor, root number, automorphy statement, target functional equation, or Hilbert–Polya operator is claimed.

Source note

Period-analysis context includes Waldvogel [1, 2] and Hsu [3]. We do not use the historical monotonicity result as a claim here and make no priority statement.

References

[1] J. Waldvogel, "The Period in the Volterra–Lotka Predator-Prey Model," *SIAM Journal on Numerical Analysis* 20(6) (1983), 1264–1272, DOI:10.1137/0720098.

[2] J. Waldvogel, "The period in the Volterra-Lotka system is monotonic," *Journal of Mathematical Analysis and Applications* 114(1) (1986), 178–184, DOI:10.1016/0022-247X(86)90076-4.

[3] S.-B. Hsu, "A Remark on the Period of the Periodic Solution in the Lotka-Volterra System," *Journal of Mathematical Analysis and Applications* 95(2) (1983), 428–436, DOI:10.1016/0022-247X(83)90117-8.

Declarations

Scope. The exact registered literal is `NO_BAD_EULER_OR_ROOT_NUMBER`. **Data and code.** The theorem, ledger, independent checks, and build instructions are released with HCS-C211. **Competing interests.** None declared. **AI-use disclosure.** Generative tools assisted drafting and code generation. The released theorem, numerical values, source metadata, and scope boundaries were checked by the internal artifact chain; this is not external peer review.